Atlantic Sea Scallop Research Track Assessment

Term of Reference 7: Research Recommendations

Research Track Peer Review Meeting
Woods Hole, MA and Webinar
April 21 – 24, 2025

TOR 7: Research Recommendations

- TOR 7. Review, evaluate, and report on the status of research recommendations from the last assessment peer review, including recommendations provided by the prior assessment working group, peer review panel, and SSC. Identify new recommendations for future research, data collection, and assessment methodology. If any ecosystem influences from TOR 1 could not be considered quantitatively under that or other TORs, describe next steps for development, testing, and review of quantitative relationships and how they could best inform assessments. Prioritize research recommendations.

Working Group's Approach to TOR 7

- Reviewed research recommendations from:
  - 2018 Research Track Assessment (SAW/SARC)
  - 2020 Management Track Assessment and Peer Review
  - Recent Science and Statistical Committee (SSC) meetings to recommend Scallop OFL and ABCs.

- Summarized the status of research recommendations from the recent stock assessments (2018 and 2020) and SSC research recommendations with respect to this research track stock assessment. Grouped by topic/theme.
- Specific response to inclusion of ecosystem influences in quantitative approaches.
- New research recommendations are categorized by topic area.

Summary of Responses: Surveys

Research Recommendation	Working Group's Response
Further investigate methods for better survey coordination between the various survey programs, including survey design, timing, and standardized data formatting for easier sharing. (2018 Benchmark Assessment)	The NEMFC and Northeast Fisheries Science Center (NEFSC) formed a Scallop Survey Working Group (SSWG) to address these topics. The effort looked at four TORs, including survey spatial coverage, sampling intensity and frequency, data standardization, storage, and access, potential impacts from the development of offshore wind, and data needs to support future stock assessments.
Investigate changes in dredge efficiency and saturation due to high scallop densities or high bycatch rates. (2018 Benchmark Assessment)	Not completed. Ongoing research at the Virginia Institute of Marine Science.

Summary of Responses:

Research Recommendation	Working Group's Response
Collect information needed for the management of the Northern Gulf of Maine (NGOM) fishery and development of appropriate reference points including biological parameters, fishery-independent surveys, and fishery-dependent data. (2018 Benchmark Assessment)	Members of the research track working group assembled a NGOM working paper, which includes updated survey and fishery data. The survey-time series continues to grow with continued Research Set-Aside support. Additional data is needed to develop reference points for this region.
Improve training of annotators used in optical surveys and develop standardized QA/QC procedures for data collected from imagery. (2018 Benchmark Assessment)	Standardized QA/QC procedures have been developed for NEFSC HabCam, implemented in 2022. SMAST Drop Camera program also uses standardized training sets, and a QA/QC process. Substantial progress has been made.
Investigate the use of software for automated annotation of imagery from optical surveys. (2018 Benchmark Assessment)	Work is underway at NEFSC. AI algorithms continue to improve rapidly. Publication on sand dollars. Staff capacity has been a limitation. Other survey groups that work with optical tools (SMAST Drop Cam and Coonamessett Farm Foundation HabCam) have ongoing projects in this area.

Summary of Responses:

Research Recommendation	Working Group's Response
Consideration of the future of surveys in the GOM region be included in the ongoing Scallop Survey Working Group (SSWG) and NEFSC-supported scallop survey re-stratification efforts. (2021 SSC Report)	A sub-group of the SSC reviewed and recommended the use of Generalized Random Tessellation Stratified (GRTS) method for survey design for Georges Bank and the Mid-Atlantic. Future implementation could be done in the Gulf of Maine. The sub-group report is available at this page: https://www.nefmc.org/library/september-2024-ssc-report. There are two recommendations from the SSWG report that are particularly relevant to surveys of the Gulf of Maine region: 1) NGOM management area and Gulf of Maine resource area should be included in regular survey coverage; 2) effort should be made to match appropriate sampling tools, designs, and methods with specific conditions of survey areas.

Summary of Responses:

Research Recommendation	Working Group's Response
Continued survey data collection and analyses of scallop populations in the Gulf of Maine to support the future development of region-specific reference points and growth estimates.	The Scallop RSA continues to support annual surveys in the Gulf of Maine. A working paper was prepared as part of this assessment to summarize the state of survey information.
Further work to develop gonad-based estimates of SSB and reference points.(2020 management track assessment)	Gonad-based estimates were explored in this research track in the Mid-Atlantic SYM model. Ultimately, the working group did not recommend transitioning to this approach because the data was very noisy - would require more work to use to develop a Mid-Atlantic shell height to gonad weight relationship.

Summary of Responses:

Research Recommendation	Working Group’s Response
Further refine and test methods for forecasting LPUE. (2018 Benchmark Assessment)	Limited progress, and further review of LPUE models is needed. Some development of spatial choice models for scallop fishermen, which have a role in LPUE forecasts. Changes to the LPUE model in annual management actions.
Develop a spatially-explicit methodology for forecasting the abundance and distribution of sea scallops by incorporating spatial data from surveys, landings, and fleet effort (aka GEOSAMS). (2018 Benchmark Assessment)	Progress has been made. Presentation on the development of GEOSAMS in January. Funding for this work was halted due to a budget cut that supported a part-time statistical programmer.
Revive and streamline previously-developed methods for interpreting VMS data. (2018 Benchmark Assessment)	Limited progress. Scallop Plan Development has developed a standard way to categorize fishing activity using speed filters. The simple speed cutoff can mask fishing and/or steaming. Using a depth cutoff would help, and the working group noted that there may be more sophisticated methods to explore, such as a Hidden Markov model.

Summary of Responses: Projections

Research Recommendation	Working Group’s Response
The SAMS model seems to be having some difficulty capturing some of the recent stock changes. Recommendation to look into new treatments for recruitment assumptions in the SAMS model. A revamp of the SAMS model to allow for more spatial estimation would be another fruitful area to explore. The SSC recommends a review of the SAMS model in the next management track assessment, and supports NEFSC’s development of a geostatistical SAMS model for the 2024 research track assessment. (2020 SSC Report)	No management track or formal review of SAMS has occurred. The working group anticipates that TOR 6 will be partially fulfilled. Problem: overestimation. In the CASA and SYM models presented by the working group, estimates of M have increased in this assessment. These changes can be carried through in the SAMS model. Downward adjustments to recruitment assumptions in SAMS may also improve performance in aggregate for the model. The working group discussed the practical importance of tracking and reviewing the performance of forecasts for individual areas, since these are used in making fishery allocations.
Research on potential drivers of changes in sea scallop stock dynamics (e.g., changing ocean conditions, including ocean acidification and warming) be included in the upcoming review of the SAMS model and in the 2024 research track assessment for scallops. (2021 SSC Report)	New sources of information include the working paper on thermal conditions (TOR 1), and revised CASA models (TOR 4). There is ongoing work at the NEFSC’s Milford Lab on ocean acidification.

Summary of Responses:

Research Recommendation	Working Group's Response
The SSC recommends continued evaluation of the performance of the projection model and the need for a more holistic evaluation of changes in stock dynamics, a synoptic evaluation of potential drivers (e.g., changing ocean conditions, including ocean warming), and revision of model assumptions to account for these changes. (2022 SSC Report)	See previous responses. Some annual ad-hoc adjustments to M were made for management purposes in estimation areas at the southern end of the range (Virginia Beach, Delmarva). A retrospective-type analysis on historical area-specific SAMS projections is recommended; a quantitative measure of past bias could be used as a tool to inform future projections. (2024 SSC Report) Summary of SAMS performance over time is shown in Figure 4.2 in TOR 6. Future work can focus on plots by sub-areas, possibly later this year.
Continued investigation of discard mortality, particularly during warm water periods, by incorporating environmental data. (2018 Benchmark Assessment)	Thermal mortality has not yet been adequately addressed - still assuming 20% discard mortality but probably too low in Mid-Atlantic because of water temperatures. Discard mortality remains an uncertainty, and further work is needed.

Summary of Responses:

Research Recommendation	Working Group's Response
Investigation of the environmental drivers affecting the productivity of this stock, such as those influencing recruitment and natural mortality. (2022 SSC Report) Research on the implications of expected future increases in bottom temperature on natural mortality and how these impacts on scallop survival will likely continue to extend northward. The SSC suggests that a thorough examination of the impacts of ecosystem and climate change on population dynamics is critical during the ongoing Sea Scallop Research Track Stock Assessment. (2023 SSC Report) The SSC supports continued monitoring of changes in the dynamics of this stock (i.e., recruitment, growth, and natural mortality) and research to understand the role of environmental drivers as this represents the most pressing concern facing the future of this fishery. (2024 SSC Report)	These three recommendations are partially addressed in TOR 1, however additional work is needed. The working group noted the importance of integrating changing environmental and oceanographic information into scallop science and management, and the need for cross-branch collaboration with the Ecosystem Dynamics and Assessment Branch (EDAB). Additional collaborative and trans-disciplinary research programs are needed to advance this work.

Summary of Responses:

Research Recommendation	Working Group’s Response
Analyze past juvenile scallop mortality events and develop better methods to model time-varying mortality in the assessment models. (2018 Benchmark Assessment)	Continued work on this topic in this research track assessment (TOR 4). The CASA models include age/time varying estimates of natural mortality.
Continue development of scallop ageing methods and examination of scallop growth processes including density dependent effects. (2018 Benchmark Assessment)	More work should be conducted to examine the estimation methods of growth transition matrices, enabling these matrices to better capture spatial and temporal changes in growth so that these changes can be incorporated into CASA models. Peer reviewed work in this area: Kowaleski et. al. (2024)

Summary of Responses:

Research Recommendation	Working Group's Response
Investigate methods to better estimating biomass and abundance variances from Hab-cam optical surveys including development of Bayesian geostatistical methods. (2018 Benchmark Assessment)	Progress has been made in this area, but it is not ready for implementation. Peer-reviewed paper on this topic: Duskey et. al. (2023).
Investigate and estimate current and historical unreported landings and effects of spatially heterogeneous fishing mortality on mortality estimates. (2018 Benchmark Assessment)	No recent progress on this topic. Unreported fishing mortality early in the assessment time series is suspected, though there are no obvious solutions to address the challenge. Hart papers in 2001 and 2003.

Summary of Responses: Mortality, Disease, Parasites

Research Recommendation	Working Group's Response
Investigate and parameterize sub-lethal effects of disease, parasites, or discarding on mortality, growth, and landings. (2018 Benchmark Assessment)	Ongoing work at the Virginia Institute of Marine Science looking at shell disease, and nematodes. Multiple survey groups are tracking scallop condition. Further exploration conducted by researchers at VIMS, see: Rudders et. al. (2019).
Continue improvements of observer recordings for vessel fishing behavior including deck loading and shucking dynamics in responses to disease or poor scallop health. (2018 Benchmark Assessment)	Observer protocols to track scallop health and meat condition (gray meats and parasites). Observers not currently checking for shell blister disease.
Continue investigating the extent of incidental fishing mortality, particularly on hard bottom habitats. (2018 Benchmark Assessment)	Not complete. Research in this area by the Virginia Institute of Marine Science and University of Delaware in 2017. This work did not focus on hard bottom habitats. See more at: Ferraro et. al. BACI study.

Summary of Responses:

Research Recommendation	Working Group's Response
Continued development of the SYM model that models selectivity dynamically as a function of full recruitment fishing mortality. (2020 Management Track)	• Not complete, and work is still underway at the NEFSC.
Transitioning the CASA model into the "next generation" of assessments would also be a natural progression; a state-space model with parameters that can be connected to environmental variables. (SSC)	• The working group recommended updating the CASA models for this research track at the start of this process in 2023. This SSC recommendation is carried forward in future research recommendations below, and in the working group's response to the TOR 7 on quantitative assessment of ecosystem influences. • CASA is attempting to capture process error in time-varying natural mortality. • State-space models are being used for this scallop species in other regions. DFO Canada is using a delayed difference state-space model to assess the Atlantic sea scallop resource.

Quantitative Assessment of Ecosystem Influences

- Ecosystem influences considered in TOR 1 were not incorporated into the CASA models.
- In 2024, the SSC recommended the consideration of a state-space assessment model.
- Other avenues for informing scientific and management decisions using environmental data in New England. The NEFMC's Risk Policy identifies climate and ecosystem considerations (climate vulnerability and fish condition) as being important in the development of management advice. These factors are evaluated and translated into a level of risk aversion for use in management and catch advice.
- The working group noted the importance of ongoing support from the NEFSC's Ecosystem Dynamics and Assessment Branch to integrate changing environmental and oceanographic information into scallop science and management.

New Recommendations

The recommendations/categories are listed below in proposed order of priority

Category	Working Group's Recommendations
Aging	• Age scallops that are available from the 1990s to fill data gaps.
Projection method	• Review the performance of area-specific forecasts (e.g. a specific SAMS estimation areas) over time. Consider the status of an area (open and being fished vs. closed) in this review. Reference earlier work complete as part of the Council's Evaluation of Rotational Management. • Develop spatial strata (SAMS Areas) in the Gulf of Maine that can be used to direct future surveys, and would be the basis for future forecasting of abundance and biomass. • Continue the development of new forecasting models, including GeoSAMS.

New Recommendations

Category	Working Group’s Recommendations
Assessment models	• Further investigate the impacts of environmental impacts on scallops, and consider ways to incorporate these directly into stock assessment models. • Develop a state-space assessment model with parameters that can be connected to environmental variables.
Growth	• Length based CASA models are dependent upon growth estimates. Research that investigates estimates of growth, and an exhaustive review of prior estimates of growth should be conducted.

New Recommendations

Category	Working Group's Recommendations
Natural Mortality	• Explore and refine predator-prey interaction models to improve natural mortality estimates. Investigate changes in M due to increases in sea star predation.
Stock Structure	• Examine current assumptions about scallop stock structure in the Gulf of Maine, Georges Bank, and the Mid-Atlantic.

New Recommendations

Category	Working Group's Recommendations
Environmental Change	• Investigate the implications of warming sea surface temperatures and the reduced optimal thermal window on larval survival to refine models of density-independent mortality. • Research to characterize the interplay of food availability, oxygen, and temperature on natural mortality. • Investigate how ocean forecasting products perform for scallop usability in assessment and forecasting models.
Survey Data	• Continue to analyze the use of artificial intelligence (AI) algorithms in annotations from images taken using optical survey tools (HabCam and Drop Camera). • Continue to survey and analyze scallops in the Gulf of Maine.